

RESEARCH ARTICLE

Discovery of Perilloxazole Pseudo-Natural Products Yields a New Sterol Biosynthesis Inhibitor Chemotype

Peng Lei^{1,2,3}  | Yuge Hu¹ | Tiantian Sun¹ | Yanyan Qu⁴ | Hanxiao Yuan¹ | Yanqing Gao^{1,2} | Jianqiang Miao^{1,3}  | Xingxing Lu^{1,3}  | Xili Liu^{1,3} 

¹College of Plant Protection, Northwest A&F University, Yangling, Shaanxi, China | ²Shaanxi Provincial Center for Biopesticide Engineering & Technology, Northwest A&F University, Yangling, Shaanxi, China | ³State Key Laboratory for Crop Stress Resistance and High-Efficiency Production, Northwest A&F University, Yangling, Shaanxi, China | ⁴Institute of Plant Protection, Beijing Academy of Agricultural and Forestry Sciences, Beijing, China

Correspondence: Peng Lei (peng.lei@nwfau.edu.cn) | Xingxing Lu (luxx@nwfau.edu.cn) | Xili Liu (seedling@nwfau.edu.cn)

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ABSTRACT

Pseudo-natural products (PNPs) are new arrangements of natural product (NP) fragments, creating chemical spaces that extend beyond those explored by nature and are unattainable via known biosynthetic pathways. Herein, we establish the design principles for agrochemically-active PNPs based on characteristic pesticide features. We systematically applied this strategy to discover PNP agricultural fungicides, opening up a broad new avenue for developing novel NP-derived pesticides. We designed perilloxazole PNPs by integrating cyclohexene and isoxazoline fragments. Cheminformatics analysis demonstrates that perilloxazole PNPs combine the advantageous properties of NPs and pesticides. Biological evaluation reveals that perilloxazole represents a new chemotype of sterol biosynthesis inhibitor with low toxicity to non-target organisms, providing a foundation for the discovery of superior green fungicides. Perilloxazole PNPs exhibit notable control efficacy against major crop diseases that cause severe agricultural losses, holding broad application prospects and high market potential.

1 | Introduction

In agricultural production, pests, including diseases, insect pests, and weeds, are the main factors seriously threatening crop harvests [1]. For example, the outbreak and epidemic of *Fusarium* head blight (FHB), a fungal disease caused by *Fusarium graminearum*, pose a serious threat to the production safety of major grains such as wheat and barley [2]. It not only leads to crop yield reduction and significant economic losses but also produces mycotoxins such as deoxynivalenol, contaminating grains and endangering the life and health of humans and animals [3]. Pesticides are indispensable materials in agricultural production. They can effectively reduce food production losses caused by pests, and contribute to ensuring food security, which is of vital importance to people around the world [4]. However, problems such as

resistance to existing pesticides and environmental pollution are becoming increasingly prominent, restricting the development of sustainable agriculture [5]. To tackle these issues, it is necessary to design and discover novel small molecule chemotypes with good biological activity and high safety [6]. One effective approach is to develop agrochemically active compounds derived from natural products. They represent a key area in the development of plant protection and are in high demand [7, 8].

Natural products (NPs) have played a role as inspiration and resources in pesticide discovery and chemical biology (Figure 1A) [7, 8]. Through traditional methods such as extraction and isolation from plants and microbial fermentation, various active NPs with unique structures can be obtained [9, 10]. However, the known NPs account for only a small part of the total NP-type

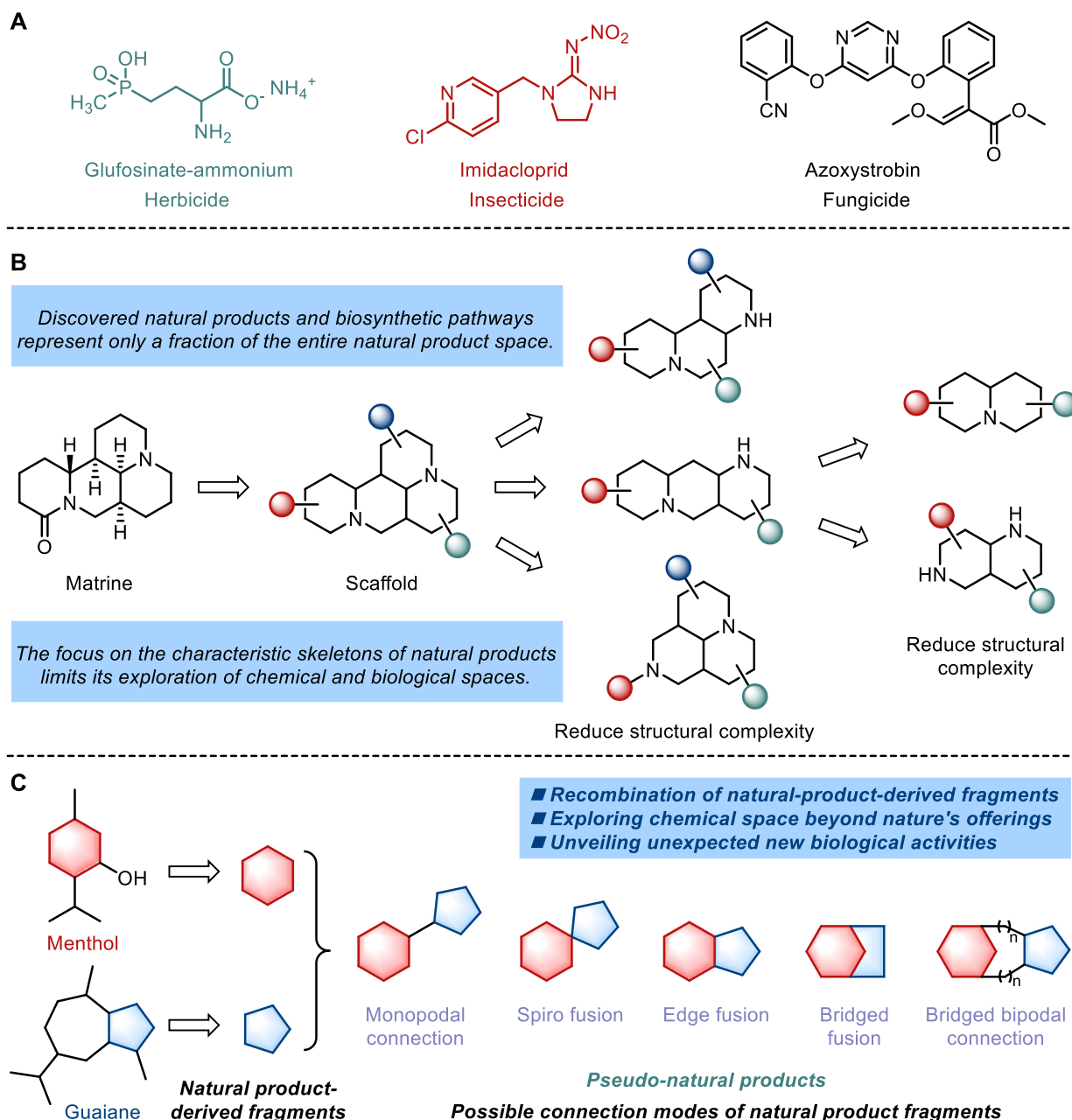


FIGURE 1 | Background and design of pseudo-natural products. (A) Natural product-derived pesticides. (B) Biology-oriented synthesis draws inspiration from NPs. (C) Design of pseudo-natural products (PNPs).

chemical space, and NP-based pesticides also represent only a minority of commercialized pesticide varieties [11]. There are still a large number of undiscovered NPs in nature. Together with the unexploited biosynthetic pathways, in principle, the inspiration from NP structures can far exceed the current scaffold of known NPs [12]. In the biology-oriented synthesis (BIOS) of NP-inspired compound classes, NP scaffolds are structurally simplified into less complex and synthesizable scaffolds that retain the characteristic properties and biological activities of the original NPs (Figure 1B) [13–15]. The focus of BIOS on the characteristic skeletons of NPs limits its exploration of chemical and biological spaces [12].

The concept of pseudo-natural products (PNPs) innovatively proposed by Waldmann et al. overcomes these limitations (Figure 1C)

[16]. PNPs are new arrangements and combinations of NP-derived fragments, generating unprecedented compound classes that cannot be obtained through known biosynthetic pathways, thus going beyond the chemical space explored by nature [17]. PNPs inherit the characteristic elements of NP structures and properties, thus having biological relevance, but may bring unexpected new biological activities [18]. The proposal of PNPs has led the fields of chemical biology and medicinal chemistry into a transformative era. The feasibility and effectiveness of this approach have been proven in the field of drug discovery [19]. For example, using this strategy, novel chemical-type PNP inhibitors of IDO1 [20, 21], osteogenesis [22], aurora kinase [23], the Hedgehog signaling pathway [24], tubulin polymerization [25], TNK kinase [26], the RHO GDP-dissociation [27], and autophagy [28] have been discovered.

However, as far as we know, PNPs have not been systematically proposed and applied in the field of pesticides. Although the development of pesticides and pharmaceuticals both belong to the field of active compound design and discovery, pesticide research and development (R&D) have unique characteristics due to the diversity of pests and different usage scenarios [29]. For example, compared with pharmaceuticals, pesticides may have significantly different characteristics in terms of absorption, distribution, metabolism, excretion, and toxicity; insecticides, fungicides, and herbicides are used to control very different species, and there are large differences in the requirements for penetrating the epidermis of target organisms; since pesticides are used outdoors, photostability is one of the most important properties to be considered for pesticide molecules [29, 30]. Therefore, it is reasonable to assume that the compositional properties of pesticides may differ from those of pharmaceuticals in terms of bioavailability [29, 31]. In addition, pesticides not only need to have high activity against pests, but also need to consider the safety of non-targets [32]. The application objects of pesticides and their own characteristics require the production process to be simple to operate and the cost to be affordable [33]. Therefore, the sustainable development of agriculture poses higher requirements and challenges for the development of new pesticides.

Herein, based on the particularities of pesticides, we newly defined the design principles of agrochemically active PNPs and systematically applied this strategy for the first time to discover PNP agricultural fungicides. The cyclohexene and isoxazoline fragments exist separately in various NPs. For example, NPs such as perillaldehyde [34, 35], dipandensin A [36], ampicordin C [37], and pandensinols C [36] contain the cyclohexene fragment. On the other hand, NPs such as acivicin [38], lacunosin A [39], phomopsis A [40], and psammaphysin A [41] all possess the isoxazoline fragment (Figure 2). However, we have not found a case where the cyclohexene and isoxazoline fragments are directly monopodally connected in NPs, which is biologically unexplored. We combined the cyclohexene and isoxazoline fragments and named it perilloxazole (Figure 2), discovering a new inhibitor chemotype acting on the ergosterol biosynthesis pathway, and creating PNP agricultural fungicides with good control efficacy against crop diseases that cause severe damage in agricultural production for the first time. Given the huge market demand, perilloxazole PNP shows promising prospects for wide-ranging applications.

There are several noteworthy features that make this research of broad interest to the global scientific community: (1) We newly defined the design principles of agrochemically active PNPs according to the characteristics of pesticides; (2) We systematically applied this strategy in pesticide discovery for the first time, opening a broad avenue for the R&D of natural-source new pesticides; (3) Using this method, we discovered an effective and safe potential fungicide for major diseases of important crops, which have broad application prospects due to high demand; (4) We discovered that the candidate fungicide acts on the sterol biosynthesis pathway but does not target CYP51, presenting a novel mechanism of action, which provides a biological basis for the discovery of novel fungicides.

2 | Results and Discussion

2.1 | Design and Synthesis of the Perilloxazole PNP Collection

Both pesticides and pharmaceuticals are bioactive compounds. Therefore, when designing agrochemically bioactive PNPs, they should meet the design principles proposed by Waldmann [16]. However, due to the specific application of pesticides in plant protection, there are different requirements for them [29]. Chirality and stereospecific content are beneficial to biological relevance. Thus, the PNP preferably has a three-dimensional scaffold [16]. Nevertheless, considering that pesticides ultimately need to be applied in agricultural production, low production cost is of great importance [33]. Therefore, when the designed pesticidal bioactive PNPs are chiral, the synthesized target compounds do not necessarily need to be single isomers, which depends on the difficulty of synthesis and the cost. This explains why most commercialized chiral pesticides in the market exist as mixtures [42]. Here, we newly define the design principles of agrochemically bioactive PNPs as follows: combine NP fragments with complementary heteroatoms, especially nitrogen and oxygen, as they are frequently encountered in pesticides and NPs; NP fragments from different organisms or biosynthetic pathways, and with different biological activities, are conducive to generating new biological activities and increasing biological relevance; for chiral PNP compounds, the synthetic target products can be either a single isomer or a mixture of isomers. The design of pesticidal-active PNPs can be guided by this design principle but should not be restricted by it. In practice, the design of target molecules can be carried out by skillfully utilizing certain aspects of this principle according to experimental or functional requirements.

Cyclohexene and isoxazoline fragments exist independently in various NPs [34–41]. Their direct monopodal connection has not been reported in NPs and remains biologically unexamined. To develop agrochemically bioactive PNPs, we combined the cyclohexene and isoxazoline fragments (Figure 2). A variety of substituents, such as benzene rings, aromatic heterocycles, and styryl groups, were introduced at the 3-position of the isoxazoline ring to further enrich the structural diversity of the target compounds. The aromatic ring contains substituents with neutral, electron-withdrawing, and electron-donating groups, which can be positioned at the ortho-, meta-, or para-positions, and includes both mono- and di-substituted patterns. Additionally, these substituents include groups with either small or large steric hindrance. The heterocycles selected included pyridine, thiophene, thiazole, furan, tetrahydrofuran, pyrrole, indole, and quinoline. An aldehyde group was introduced onto the cyclohexene. Subsequently, based on this modification, substituted hydrazone, acylhydrazone, and sulfonylhydrazone groups were incorporated via further reactions. Moreover, several cyclohexene structures bearing other substituents were introduced, including alcohol, methyl, and ketone carbonyl groups, as well as unsubstituted cyclohexene structures. Considering the specific needs of agricultural pesticides, we selected a simple-to-operate and cost-effective synthetic route (Figure 3A). First, aromatic aldehydes and hydroxylamine hydrochloride undergo a condensation reaction to form the corresponding oximes. Then, chlorination

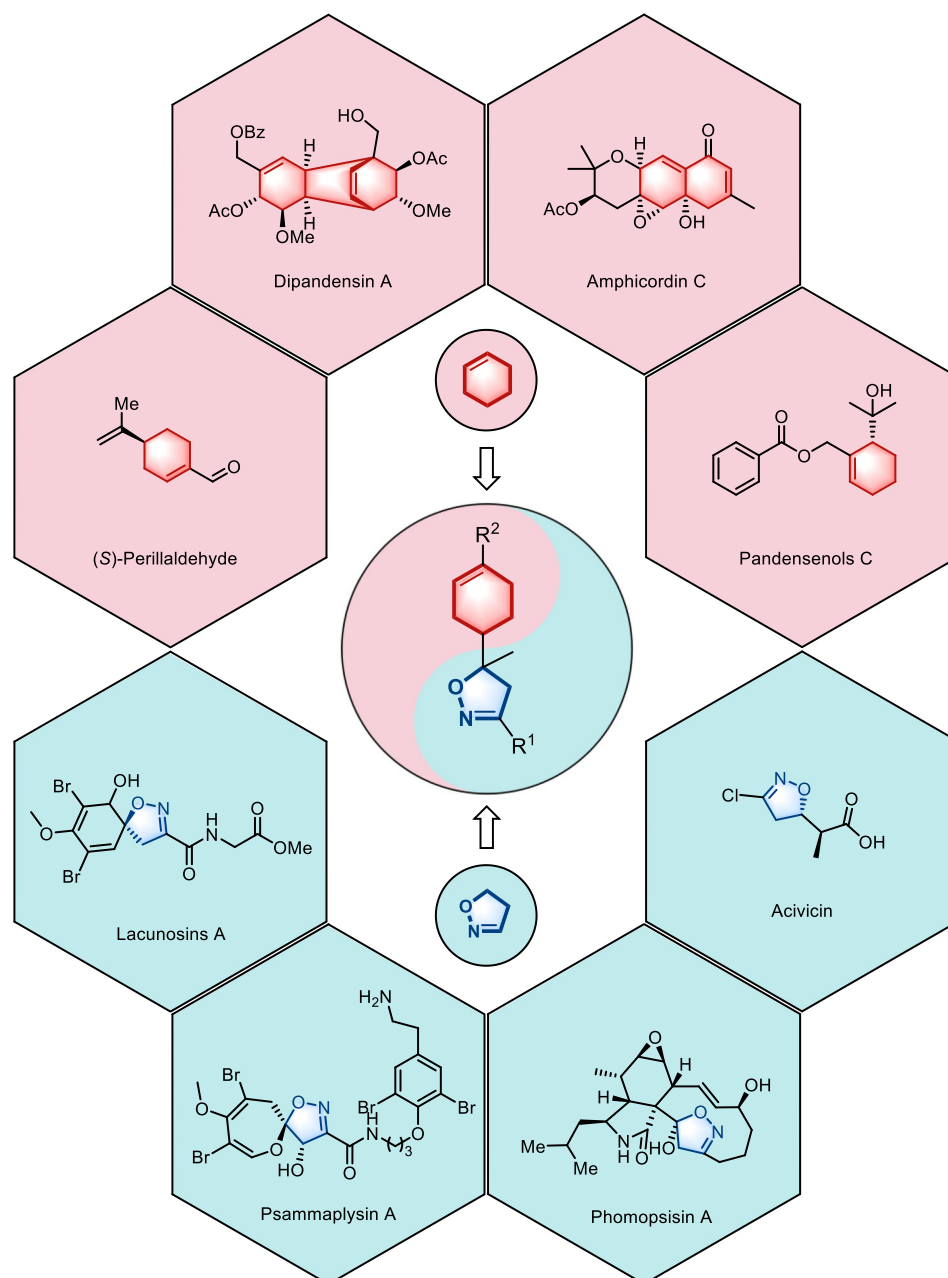


FIGURE 2 | Design strategy of the perilloxazole pseudo-natural product. Cyclohexene natural products (light pink hexagons), isoxazoline natural products (light blue hexagons), and the direct monopodal connection between cyclohexene and isoxazoline fragments (circles).

is carried out under the action of *N*-chlorosuccinimide to obtain α -chloro-oximes. Subsequently, dehydrohalogenation occurs to form dipolar nitrile oxides, which then undergo a 1,3-dipolar cycloaddition reaction with perillaldehyde to construct the isoxazoline ring, thereby preparing the target compound **5**. On this basis, the reaction of compound **5f** with hydrazine, acylhydrazine, or sulfonylhydrazine can produce the target compound **6**. Replacement of perillaldehyde used in the synthesis of compound **5** with other substituted 4-vinylcyclohex-1-ene can yield compound **7**. The above-mentioned synthetic method has the characteristics of readily available and inexpensive raw materials, mild reaction conditions, and simple operation. We first optimized the synthesis conditions of the target compound **5** (Figure 3B). We investigated the effects of stoichiometry, solvent, and base. After extensive optimization, we found that

the optimized conditions involve a slight excess of perillaldehyde and sodium carbonate in DCM at room temperature. With the optimized conditions in hand, a total of 74 PNPs combining cyclohexene and isoxazoline fragments with diverse structures were efficiently obtained in a convenient manner (Figure 4). Among them, over 30 g of compound **5f** was prepared (Figure 3B). It should be noted that this method lacks stereoselectivity. Since the perillaldehyde we used is itself an *S*-isomer, it is speculated that PNP **5f** are diastereoisomers.

The stability of compounds is a crucial factor in the biological activity testing of pesticides. During the agricultural antifungal activity assay, the compound solution needs to be incorporated into the culture medium at approximately 50°C, followed by incubation in the dark at around 25°C for several dozen hours.

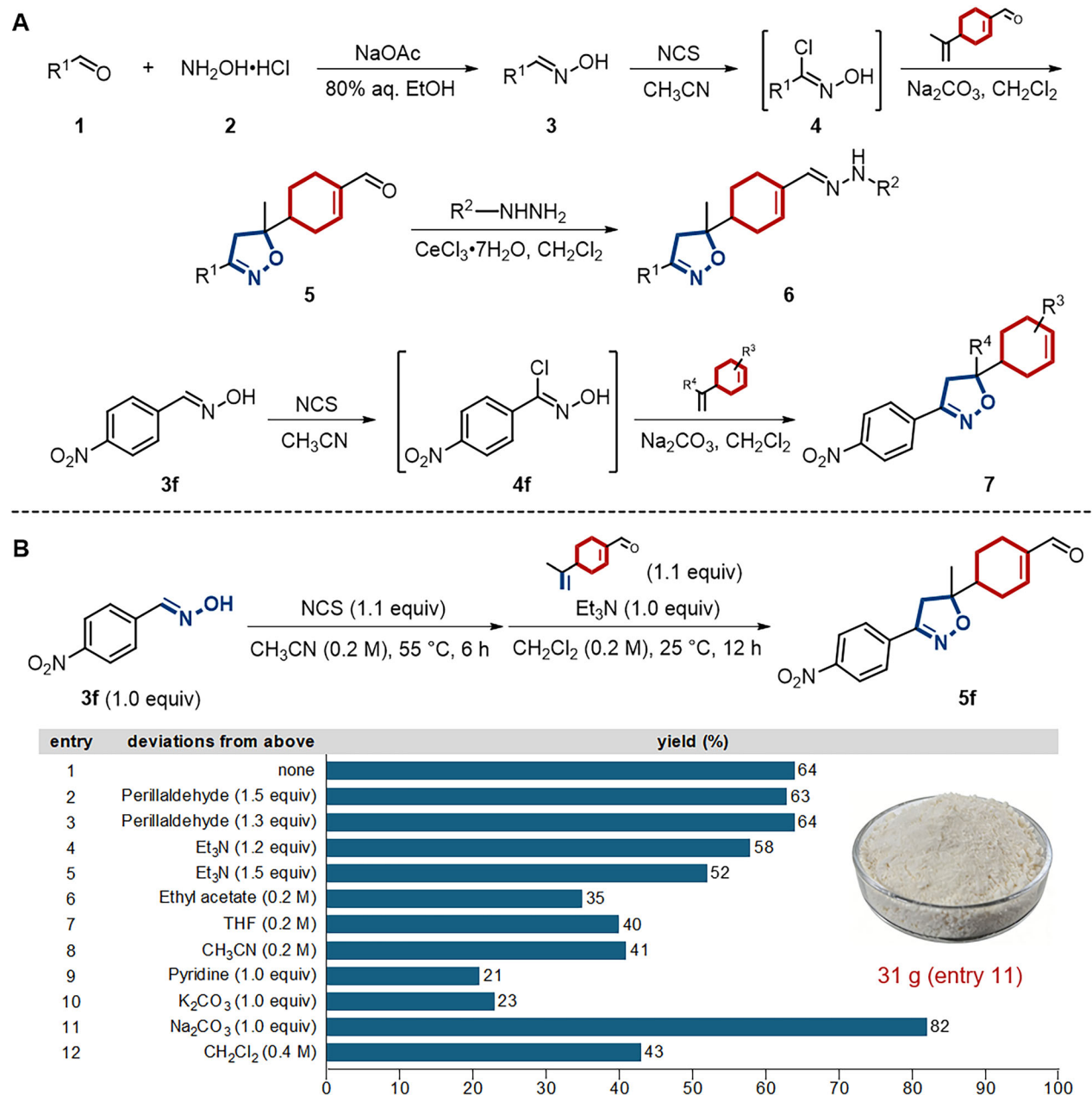


FIGURE 3 | Synthesis of the perilloxazole PNPs. (A) Synthetic routes of perilloxazole PNPs. (B) Optimization of the conditions.

Consequently, we selected **5f** from aldehydes, **6a** from hydrazones, and **6l** from acylhydrazones, respectively, to assess their stability after exposure to 50 °C for 72 h. The results indicated that no substantial changes occurred in the synthesized aldehyde, hydrazone, and acylhydrazone perilloxazole PNPs after the treatment (Figure S1).

2.2 | Cheminformatics Analysis

The structures, physicochemical properties, NP likeness, and pesticide likeness of the target compounds were compared. The NP-likeness score proposed by Ertl et al. ranges from -5.0 to +5.0 [43]. The larger the value, the more it conforms to the characteristics of NPs. The NP-likeness scores of the designed perilloxazole PNPs range from -0.5 to +1.5, with the majority

distributed between +0.7 and +1.1. The score distribution is narrow in the distinction sparsely covered by NPs and drugs; that is, they are between the two groups (Figure S2A) [16]. The shape diversity of the PNP collection was evaluated by calculating the three principal moments of inertia (PMI) of the molecules [44]. The designed PNPs disperse from rod- to disc- and sphere-like shapes. However, in order to obtain perilloxazole PNPs with more diverse shapes, in-depth structural design is still in progress (Figure S2B). Pesticide likeness, like pharmaceutical likeness, has specific requirements for molecular weight (MW), LogP, hydrogen-bond acceptors (HBA), hydrogen-bond donors (HBD), and rotatable bonds (ROB). In addition, since pesticides are used for plant protection in the natural environment, there are certain requirements for their photodegradation. Therefore, the number of aromatic bonds (ARB) they contain should not exceed 17 [29]. The pesticide likeness analysis shows that most

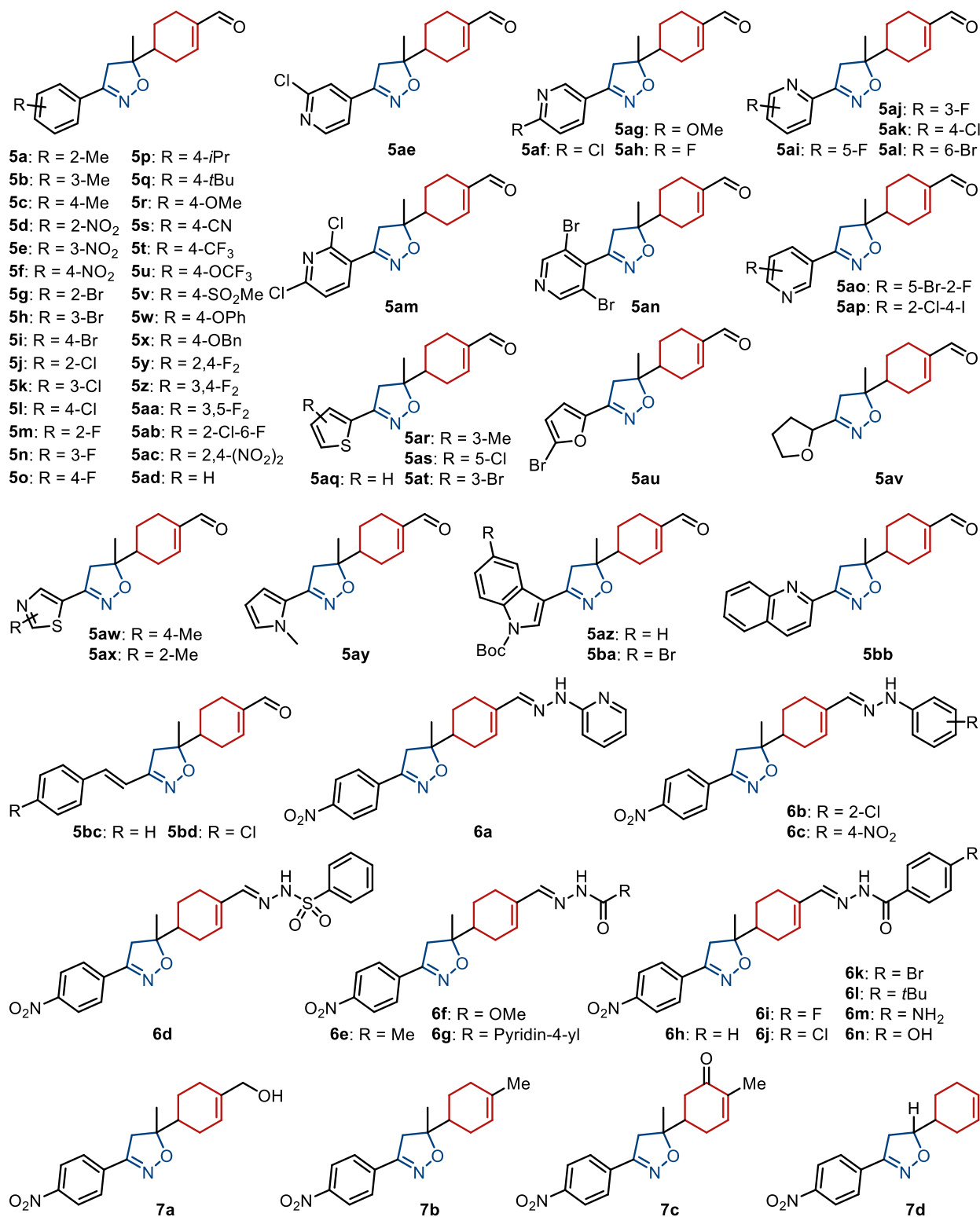


FIGURE 4 | Structures of the perilloxazole PNPs.

of the designed perilloxazole PNPs fall within the range of the physicochemical properties of pesticides (Figure S2C,D and Table S1). Cheminformatics analysis shows that the perilloxazole PNPs represented by **5f** are at the interface of the physicochemical characteristics of pesticides and NPs and can combine their favorable properties.

2.3 | Biological Activity

The combination of different NP fragments generates new biological activities, action modes, and/or targets, while losing the inherent biological activities of single fragments or fragment-sized NPs. Although using unbiased cell-based assays is an

ideal approach for pharmaceutical activity evaluation [45], to our knowledge, there is currently no such method for pesticide testing. Therefore, in accordance with the characteristics of agricultural diseases, we selectively conducted antifungal activity tests on representative pathogenic fungi. The important pathogenic fungi selected are as follows: (1) for major food crops: *Fusarium graminearum*, *Gaeumannomyces graminis* var. *tritici*, and *Rhizoctonia solani*; (2) for major oil crops: *Sclerotinia sclerotiorum*; (3) for major fruit trees: *Valsa mali*; (4) for major vegetables: *Botrytis cinerea* and *Colletotrichum orbiculare*.

The biological activity result (Figure 5A) shows that all designed PNPs exhibit certain inhibitory activities against the seven tested pathogenic fungi, with particularly good activities against *V. mali*, *B. cinerea*, *G. graminis* var. *tritici*, and *F. graminearum*. The preliminary structure-activity relationship analysis shows that for compound **5**, when the cyclohexene moiety is derived from perillaldehyde, the substituent R¹ at the 3-position of the isoxazoline ring has a significant impact on the activity. When R¹ is a substituted benzene ring, the activity is better than when it is a heterocyclic ring (**5a–5ad** > **5ae–5bb**). When R¹ is a substituted benzene ring, electron-withdrawing substituents are more conducive to the improvement of activity compared with electron-donating substituents (e.g., **5d–5o**), but the introduction of a methylsulfonyl group is unfavorable for activity (**5v**); for electron-withdrawing substituents, the activity of para-substitution is better than that of ortho- and meta-substitutions (e.g., **5f** > **5d** and **5e**), and double-substitution does not improve the antifungal activity (**5y–5aa** vs. **5m–5o**; **5ac** vs. **5f**). When R¹ is a substituted heterocyclic ring, the electronic effect of the substituents on the pyridine ring is similar to that of the benzene ring, and the activity of electron-withdrawing substituents is better than that of electron-donating substituents (**5af** > **5ag**), and double-substitution does not improve the activity (**5am–5ap**); among the five-membered heterocycles, the activities of pyrrole and thiophene are better than those of furan and thiazole (**5ay** and **5aq–5at** > **5au** and **5aw–5x**), the introduction of an alicyclic heterocycle is unfavorable for activity (**5av** < **5au**), and when the five-membered heterocycle is fused with a benzene ring, it is also unfavorable for activity (**5az–5ba** < **5ay**). Among the benzo-heterocycles, quinoline is more beneficial for the improvement of activity compared with indole (**5bb** > **5az–5ba**). Given the diverse and excellent biological activities of hydrazone compounds [46], we introduced hydrazone, acylhydrazone, and sulfonylhydrazone fragments based on compound **5f**. Among them, the activity of hydrazones is slightly better than that of acylhydrazones and sulfonylhydrazones. In the hydrazone structure, a benzene ring with chlorine or nitro substitution is more beneficial for enhancing the activity against *G. graminis* var. *tritici* compared with the heterocyclic pyridine (**6b** and **6c**); in the acylhydrazone structure, the introduction of NH₂ and OH on the benzene ring shows better activity against *G. graminis* var. *tritici* and *V. mali*. However, compared with compound **5f**, the antifungal activity of hydrazone compounds generally did not increase as expected. Instead, it led to a significant decrease, indicating that the simple -CHO substitution on cyclohexene is more favorable for activity than -C=N-NH-R². In addition, we also introduced some other cyclohexene-containing fragments. Results indicated that the antifungal activity of the -CHO group is superior to that of the -CH₂OH group (**7a**). Moreover, it is significantly more potent than the methyl group introduced by limonene (**7b**), the

methyl and carbonyl groups introduced by carvone (**7c**), and the unsubstituted case (**7d**). Finally, it is found that PNP **5f** shows the best comprehensive performance. At a concentration of 50 mg/L, it exhibits moderate to excellent inhibitory activities against the 7 tested pathogenic fungi, especially showing an inhibition rate of not less than 85% against *F. graminearum*, *G. graminis* var. *tritici*, *V. mali*, and *B. cinerea*, with EC₅₀ values of 1.15, 0.27, 0.47, and 1.29 mg/L, respectively.

PNP **5f** was selected to evaluate its protective and curative effects in plant tissues against representative and economically important fungal pathogens in agricultural production (Figure 5B–G and Tables S2–S7). As one of the world's most essential cereal crops, wheat is severely impacted by FHB [1–3]. PNP **5f** demonstrated FHB control efficacy comparable or superior to that of the commercial fungicide triadimefon, particularly at lower concentrations (Figure 5B and Table S2). Apple *Valsa* canker, caused by *V. mali*, threatens both tree health and fruit quality, affecting production and postharvest storage [47]. PNP **5f** exhibited strong protective and curative activity on apple branches (Figure 5C and Table S3) and fruits (Figure 5D and Table S4). It performed particularly well on fruits, with both protective and curative efficacy exceeding 90% at a concentration of 200 mg/L. *B. cinerea* ranks second among the top 10 fungal pathogens, infecting over 200 plant species and causing substantial pre- and postharvest losses [48]. Compared to triadimefon, PNP **5f** showed enhanced protective and curative activity on grape fruits (Figure 5E and Table S5), tomato fruits (Figure 5F and Table S6), and tomato leaves (Figure 5G and Table S7). In particular, it exhibited an outstanding effect on gray mold of tomato fruits, almost achieving complete control. PNP **5f** displayed satisfactory control effects of major crop diseases caused by *F. graminearum*, *V. mali*, and *B. cinerea*. Its activity remained robust even at reduced concentrations, outperforming triadimefon. Owing to its simple structure, straightforward synthesis, low cost, and compelling efficacy against key agricultural pathogens, this PNP holds promising application prospects and is expected to meet broad market demand.

2.4 | Toxicity Evaluation

In the R&D of agrochemically bioactive compounds, toxicity tests on mice and bees are crucial for evaluating their safety and environmental compatibility. These tests have important scientific significance and practical application value [49, 50].

Mice, as a mammalian model, have physiological and metabolic systems with a high degree of homology to those of humans and other higher mammals. Toxicity data on mice can effectively reflect the potential harm of compounds to mammals and are a core indicator for evaluating whether a compound has the potential for further development as a pesticide product [49]. Adult BALB/c mice were selected and acclimatized for 1 week prior to the experiment to ensure their healthy status. Before administering the dose, the mice were fasted overnight, while having unrestricted access to water. Subsequently, the mice were randomly grouped. The concentration of PNP **5f** was set at 1000 mg/kg, and the gavage volume was 1 mL/100 g.bw. After continuous observation for 14 days, no toxic manifestations or signs of death were observed in mice, indicating that **5f** has low

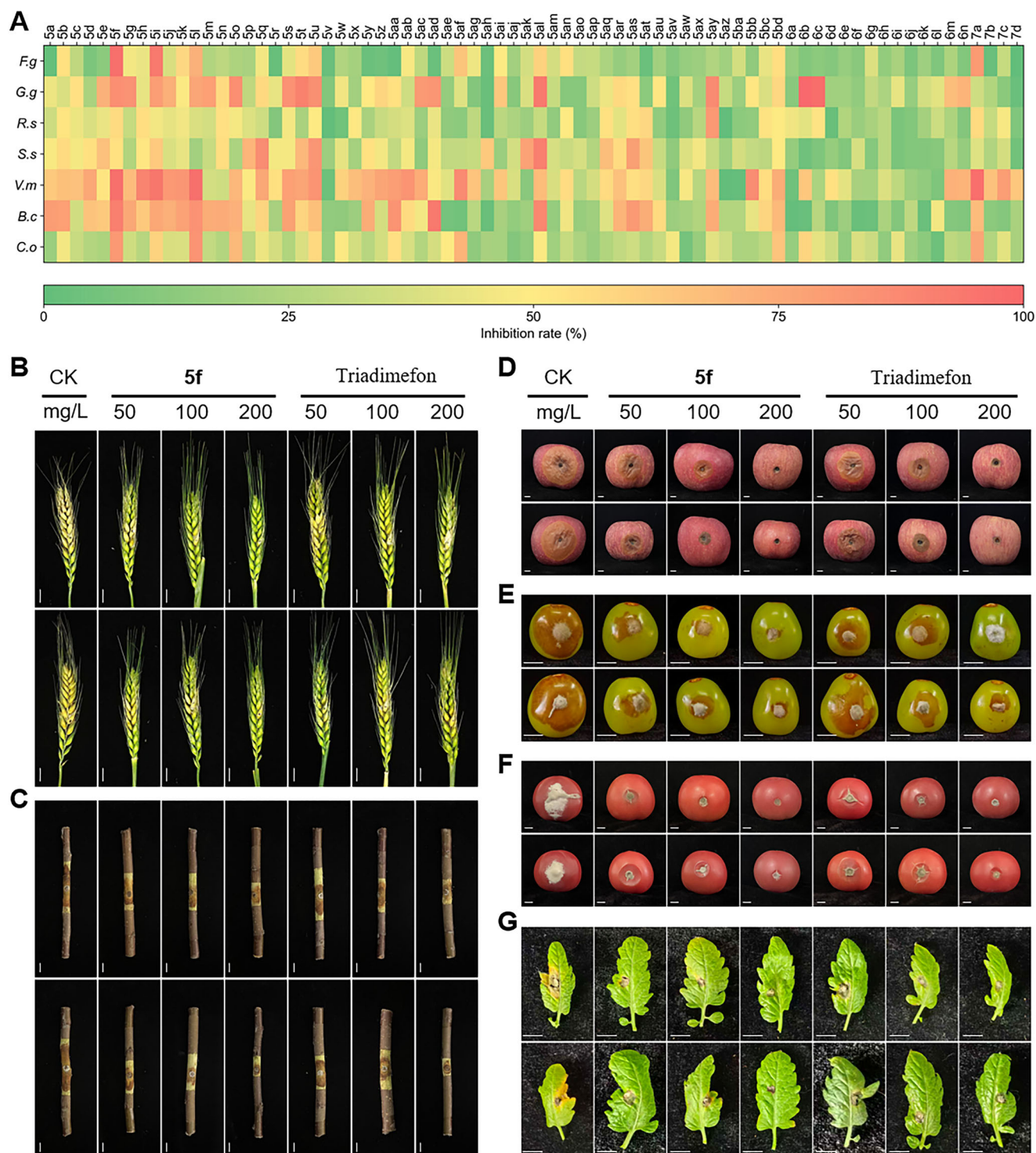


FIGURE 5 | Bioactivity of the perilloxazole PNPs. (A) In vitro antifungal activity against seven plant pathogenic fungi at 50 mg/L: *F. g.*, *Fusarium graminearum*; *G. g.*, *Gaeumannomyces graminis* var. *tritici*; *R. s.*, *Rhizoctonia solani*; *S. s.*, *Sclerotinia sclerotiorum*; *V. m.*, *Valsa mali*; *B. c.*, *Botrytis cinerea*; *C. o.*, *Colletotrichum orbiculare*. (B) Protective (top) and curative (bottom) effects against *Fusarium* head blight on wheat spikes. (C and D) Protective (top) and curative (bottom) effects against apple *Valsa* canker on apple branches and fruits, respectively. (E–G) Protective (top) and curative (bottom) effects against gray mold on grape fruits, tomato fruits, and tomato leaves, respectively. Note: For panels C, E, F and G, the corresponding compounds and their concentrations in each panel are identical to those labeled in the upper panels B and D. Scale bar: 1 cm.

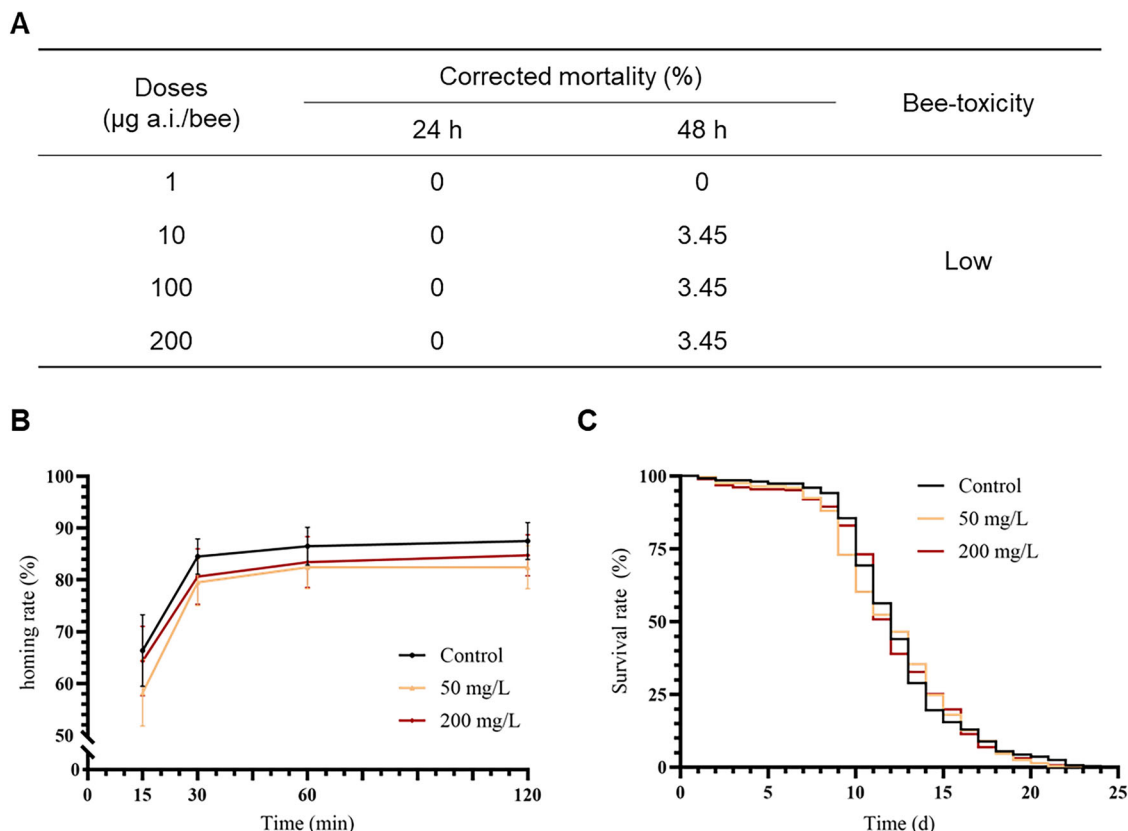


FIGURE 6 | Toxicity results of the perillioxazole **5f**. (A) Acute contact toxicity of PNP **5f** to honeybee *Apis mellifera ligustica*. High toxicity, median lethal dose (LD_{50}) < 2 $\mu\text{g/bee}$; moderate toxicity, $2 \leq \text{LD}_{50} \leq 11 \mu\text{g/bee}$; low toxicity, $\text{LD}_{50} > 11 \mu\text{g/bee}$ (US Environmental Protection Agency, 2017). (B) Impact of PNP **5f** on the homing rate of *A. mellifera ligustica* foragers. (C) Chronic toxicity assessment of PNP **5f** on *A. mellifera ligustica* survival.

toxicity to mammals [51]. The specific experimental methods are provided in the supporting information.

Honeybees, as crucial pollinators, play an indispensable role in maintaining ecosystem balance and supporting agricultural production [50]. They also serve as sensitive indicator organisms for environmental toxicity [52]. The toxicity of a compound to honeybees is directly linked to its environmental safety. If a compound is highly toxic to bees, it could potentially lead to a decline in the pollinator population, disrupt the ecological chain's stability, and even precipitate regional ecological risks [53]. Acute toxicity tests on *Apis mellifera ligustica* workers showed that the LD_{50} of compound **5f** in 48 h exceeded 200 $\mu\text{g a.i./bee}$, well above the threshold for low-toxicity pesticides ($\text{LD}_{50} > 11 \mu\text{g a.i./bee}$) (Figure 6A) [54]. Exposure to 50 and 200 mg/L did not significantly affect the homing rate of honeybee foragers within 2 h ($F = 0.360$, $p = 0.708$) (Figure 6B), nor did it shorten the lifespan of newly emerged honeybee adults ($\chi^2 = 0.515$, $p = 0.773$) (Figure 6C and Table S8). These results indicate that **5f** exhibits acute low toxicity and does not produce detectable chronic effects on honeybees. When applied as a fungicide on crops, the present results indicate that the PNP **5f** is unlikely to substantially interfere with the key neurophysiological processes or development underlying honeybee orientation, flight, and colony maintenance, at least within the range of environmentally relevant concentrations. Such target selectivity is particularly desirable in integrated pest and pollinator management (IPPM) frameworks, where protecting pollinator fitness and behavior is

essential for maintaining both crop productivity and ecosystem resilience [55].

2.5 | Mechanism of Action

To assess the mechanism of action of **5f**, a global transcriptional analysis was conducted using RNA-seq. Hierarchical clustering analysis of all expressed genes revealed a clear separation between the **5f**-treated samples and the control group (Figure S3A), indicating a pronounced transcriptional reprogramming induced by **5f**. Differential expression analysis identified a total of 1555 differentially expressed genes (DEGs), including 816 up-regulated and 739 down-regulated genes (Figure S3B).

KEGG pathway enrichment analysis showed that these DEGs were significantly enriched in pathways related to lipid and isoprenoid metabolism, redox-associated processes, and transmembrane transport (Figure S3C). Consistently, GO enrichment analysis revealed that the DEGs were predominantly associated with oxidoreductase activity, iron ion binding, heme binding, and tetrapyrrole binding, as well as transporter activity, transmembrane transporter activity, and transition metal ion transmembrane transporter activity (Figure S3D), suggesting that **5f** treatment substantially perturbs heme-dependent redox systems and membrane-associated metabolic processes in *F. graminearum*.

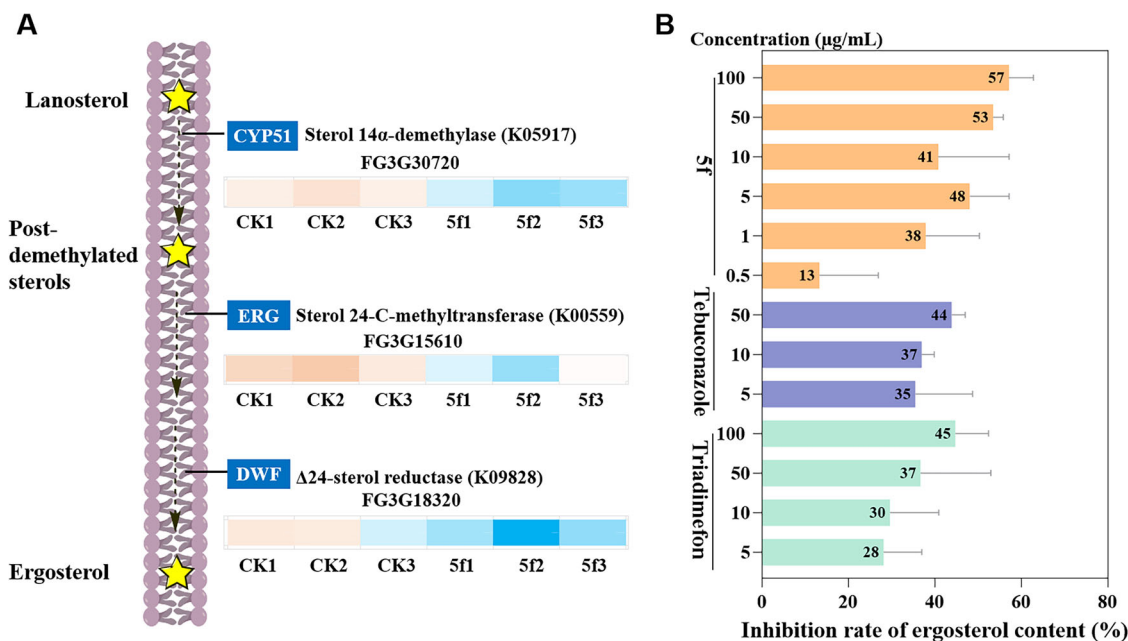


FIGURE 7 | Mechanistic study of compound **5f** against *F. graminearum*. (A) Effects of PNP **5f** on the sterol biosynthesis pathway and heatmap of differentially expressed genes. (B) Effects of PNP **5f**, triadimefon, and tebuconazole on ergosterol content.

Given that iron- and heme-dependent redox enzymes play essential roles in multiple oxidative steps of sterol biosynthesis, these enrichment patterns suggested a potential impact of **5f** on sterol-related metabolic processes. Accordingly, transcriptional changes within the sterol biosynthesis pathway were further examined (Figure 7A). Several key enzymes involved in post-lanosterol sterol modification, including sterol 14 α -demethylase (CYP51), sterol 24-C-methyltransferase (ERG), and Δ 24-sterol reductase (DWF), exhibit significant differences.

Consistent with these transcriptional changes, ergosterol content determination revealed that compound **5f** significantly interfered with sterol accumulation in *F. graminearum* (Figure 7B). Inhibition rate analysis showed that **5f** reduced ergosterol levels in a dose-dependent manner, exhibiting an inhibition efficiency comparable to that of commercial sterol biosynthesis inhibitors triadimefon and tebuconazole, indicating that **5f** effectively disrupts sterol homeostasis in fungal cells.

Given that CYP51 represents a well-established molecular target of commercial sterol biosynthesis-inhibiting fungicides such as triadimefon and tebuconazole, CYP51 was selected as a representative target for biophysical validation. CYP51A, CYP51B, and CYP51C were subjected to heterologous expression and purification, among which only CYP51B was successfully obtained. Surface plasmon resonance (SPR) assays showed that **5f** exhibited only weak binding affinity toward CYP51B, whereas triadimefon and tebuconazole displayed stronger interactions with CYP51B, with dissociation constants (K_d) of 4.55×10^{-5} and 3.34×10^{-5} M, respectively (Figure S4A). These results suggest that CYP51B is unlikely to be the primary direct target of **5f**.

Taken together, these results indicate that although **5f** clearly disrupts ergosterol biosynthesis, its mechanism of action is distinct from that of conventional CYP51-targeting fungicides. Instead, **5f** appears to affect sterol metabolism through a broader

or alternative regulatory mechanism, highlighting its potential as a novel antifungal scaffold with a non-classical mode of action.

3 | Discussion and Conclusion

Food security is of utmost importance to humanity. Pesticides can effectively reduce food losses caused by pests by controlling them. However, the long-term, large-scale, and irrational use of pesticides has led to environmental hazards and an increase in resistance levels. There is an urgent need to develop novel green pesticide molecules, which is in line with the global food security target strategy. Obtaining bioactive compounds from natural sources is a common and effective approach in modern pesticide R&D. However, current NP-based pesticide creation methods, such as BIOS, still have limitations in exploring chemical and biological spaces [12, 13]. Therefore, it was necessary and urgent to innovate new strategies for NP-based pesticide R&D.

Here, for the first time, we systematically applied the PNP strategy to discover agricultural fungicides and newly defined the design principles of agrochemically active PNPs according to the characteristics of pesticides. We designed and synthesized the perilloxazole PNP collection by combining the biosynthetically unrelated NP fragments cyclohexene and isoxazoline through direct monopodal connection. We also elucidated the chemoinformatics analysis, antifungal activity, safety assessment, and mechanism of action of this PNP collection. Chemoinformatics analysis indicated that the perilloxazole PNP collection could combine the favorable properties of NPs and pesticides. Based on the characteristics of agricultural diseases and the current situation of pesticide activity testing, we selectively chose important pathogenic fungi for biological evaluation. The analysis of antifungal activity revealed that the perilloxazole PNP collection had good inhibitory activity against plant pathogenic fungi such as *F. graminearum*, *G. graminis* var. *tritici*, *B. cinerea*, and *V.*

mali. In particular, PNP **5f** had good protective and curative effects against *Fusarium* head blight, apple *Valsa* canker, and gray mold, which cause significant damage in agricultural production. Toxicity tests on mice and bees demonstrated good safety and environmental compatibility.

Ergosterol is a critical structural component of fungal cell membranes, and disruption of its biosynthesis compromises membrane integrity, permeability, and overall cellular viability [56]. In this study, both transcriptomic analysis and ergosterol quantification consistently demonstrated that **5f** profoundly affects sterol biosynthesis in *F. graminearum*. The dysregulated expression of multiple post-lanosterol modification enzymes, coupled with the marked reduction in ergosterol content, indicates that sterol metabolism constitutes a key downstream process impacted by **5f** treatment. However, despite the clear suppression of ergosterol accumulation, SPR analysis revealed that **5f** binds only weakly to CYP51B, in contrast to classical azole fungicides, which exhibit strong and specific interactions with CYP51B. This discrepancy suggests that **5f** does not act as a typical CYP51 inhibitor and may exert a novel mechanism of action. Consistent with our transcriptome analysis, the observed ergosterol depletion is more likely attributable to indirect regulatory effects on sterol biosynthesis, potentially through perturbation of heme-dependent redox systems, membrane-associated metabolic networks, or the coordinated regulation of multiple sterol-modifying enzymes.

Previous studies have shown that certain lysosomotropic compounds can accumulate in lysosomes and disrupt intracellular lipid and sterol trafficking, thereby indirectly affecting steroid biosynthesis [57]. Given that our compound **5f** was found to interfere with ergosterol biosynthesis, we sought to determine whether this effect could arise from lysosomotropic activity. As reported, lysosomotropic compounds typically display $\text{ClogP} > 2$ and $\text{pKa} > 6.5$ [57]. Compound **5f** exhibits a ClogP value of 3.4 but a pKa of 2.9, which is substantially below the threshold for lysosomal trapping. These results suggest that **5f** is unlikely to accumulate in lysosomes or act as a lysosomotropic compound.

Furthermore, we performed relevant subcellular localization and lysosomal accumulation assays in *F. graminearum* using chloroquine, a classic lysosomotropic compound, as the positive control and dimethyl sulfoxide (DMSO) as the negative control [57]. Intracellular lipid droplets were analyzed for distribution and fluorescence intensity using the BODIPY lipid-specific red fluorescent dye. Combined with the Lyso-Tracker green fluorescent probe targeting lysosomes, dual-color colocalization analysis was conducted via laser confocal fluorescence microscopy. The results showed that mycelia treated with chloroquine exhibited enlarged lysosomes, enhanced red fluorescence, and substantial colocalization of lipids and lysosomes. In contrast, treatment with **5f** at 0.5 or 5 $\mu\text{g/mL}$ caused no significant changes in lysosomal size or red fluorescence intensity compared with the DMSO negative control, with similar colocalization patterns of lipids and lysosomes (Figure S4B). These findings demonstrate that **5f** does not interfere with ergosterol biosynthesis via lysosomal accumulation.

In summary, the PNP **5f** features a simple structure, inexpensive and easily accessible raw materials, a straightforward preparation process, low cost, a novel mechanism of action, excellent antifun-

gal activity against important pathogenic fungi, and high safety. These characteristics endow it with broad market prospects, great development value, and application feasibility. The application of the PNP strategy can not only effectively expand the chemical space of NP-like substances but also serves as an effective new tool for designing pesticide molecules, opening a new and broad avenue for the R&D of natural-source pesticides. This helps to alleviate the pressure on food demand brought about by global population growth and contributes to the innovation of pesticide varieties in the field of plant protection.

Author Contributions

Peng Lei: conceptualization, methodology, writing – original draft, writing – review and editing, supervision, project administration, funding acquisition, visualization. **Yuge Hu:** investigation, validation, data curation, formal analysis, visualization. **Tiantian Sun:** investigation, data curation, visualization. **Yanyan Qu:** investigation, visualization. **Hanxiao Yuan:** investigation. **Yanqing Gao:** visualization. **Jianqiang Miao:** visualization. **Xingxing Lu:** methodology, investigation, supervision, writing – original draft, visualization. **Xili Liu:** conceptualization, methodology, supervision, writing – original draft, writing – review and editing, funding acquisition, resources, project administration.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that supports the findings of this study are available in the Supporting Information of this article.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File 1: anie72424-sup-0001-SuppMat.pdf.